FROM FUEL TO FIBRE

India's Textile → MMF → Petrochemical Loop

How rising man-made-fibre demand and ethanol-for-petrol substitution converge to build a domestic feedstock-to-value-added-fibre chain.

Fuel → Chemicals → Fibre → Recycle → Fibre

BACKGROUND

India's textile trade at a glance

$37 bn
Textiles & apparel exports, FY25 (6th largest globally)

3.9%
India's share of global textile & apparel trade

+$532 M
Technical-textile net trade SURPLUS, FY24 (from deficit in FY19)

~$4.5 bn
MMF textiles India still IMPORTS / yr (China >55%)

Three export baskets, three fibre profiles

Apparel (HS 61 & 62)
≈ $16 bn — large surplus

Cotton-skewed; MMF lines underweight

Technical textiles (207 codes)
Exp $2.59 bn / Imp $2.05 bn

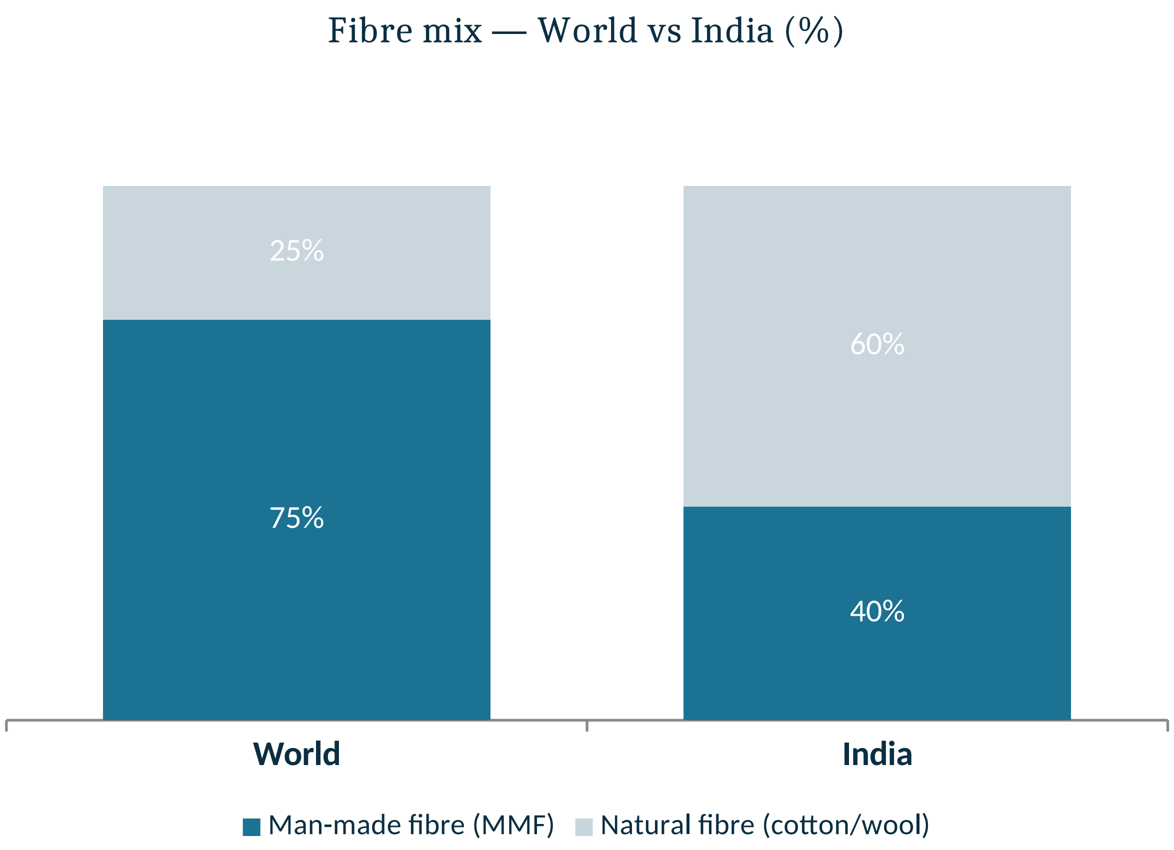
MMF + specialty-fibre intensive

Handloom (HS 50/57/58/63)
Shrinking ~ −4%/yr

Natural, artisanal, home-textile led

THE DEMAND SHIFT

The world runs on man-made fibre — India is catching up



132 MT global fibre output 2024, +6.5% YoY; synthetics now 69% (up 9pts since 2020)

78 MT polyester alone — 59% of all fibre, but 88% fossil-based

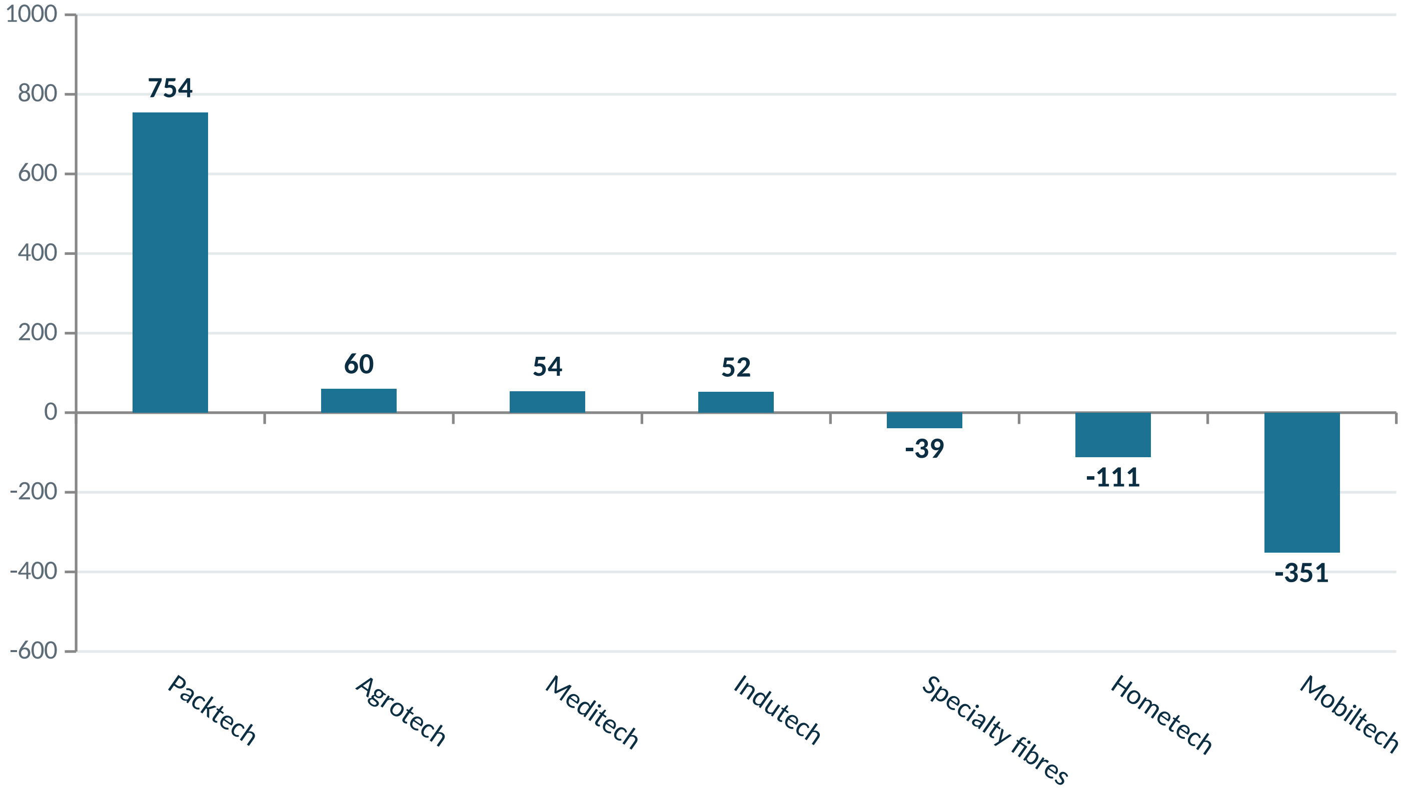
60% India's target MMF share (National Fibre Mission) — up from a cotton-heavy ~40%

PLI + NTTM incentives deliberately steer to MMF & technical textiles — not cotton

THE IMPORT GAP

India's deficit is a specialty-MMF deficit

Technical-textile net trade by segment, FY24 (USD Mn) — the surplus is Packtech-only; the deficits are MMF/specialty-fibre segments.



Net IMPORTER — the priority

- **Mobiltech −$351M**

nylon tyre cord, airbag & seatbelt yarn

- Hometech −$111M · Specialty fibres +275% imports

Net EXPORTER — the concentration risk

Packtech alone (+$754M) carries India's entire technical-textile surplus — and it's the lowest-tech, jute/PP-sack segment. Strip it out and India is still a net importer.

POLICY ANCHOR

The mandate: National Fibre Mission 2030-31

India's official target set — and it explicitly calls for the petrochemical-feedstock + recycling loop.

₹13.6 L cr

MMF's contribution to the economy, 2024-25

75 → 130

lakh MT — MMF production target by 2030-31 (CAGR 9.6%); global rank 2 → 1

24 → 31.5

lakh MT — MMF imports rising to 2030-31: the substitution prize

14.6%

India's target share of global MMF production (from 10%); China alone >70%

The gap PLI leaves open

PLI for Textiles (₹10,683 cr) boosts MMF fabric & apparel — but “focuses on downstream”. Upstream MMF feedstock & specialty fibres stay import-reliant.

The Mission's own words

“Incentivise local MMF feedstock production through sustainable chemical pathways and recycled raw material use.”

“...promoting recycled MMF production and investment in advanced petrochemical value chains.”

THE UNLOCK

Ethanol substitutes petrol — and frees the feedstock

Gasoline is a commoditised, declining “red ocean.” Petrochemicals are the growing “blue ocean.” Ethanol + EVs cap petrol demand and release naphtha & reformate to redeploy.

RED OCEAN

Petrol / gasoline

- Commoditised, thin margin, structurally declining (EVs, efficiency)
- E20 ≈ 10.8 bn L · E30 ≈ 16 bn L of petrol displaced
- Frees petrol-range naphtha + reformate

BLUE OCEAN

Petrochemicals

- Growing ~4.25%/yr in Asia; ~90% of world crude-demand growth is petrochemical-driven
- Integrated refiner adds ~$0.7–2.0 / bbl vs a fuels-only peer
- The reformer that makes petrol octane makes paraxylene

FEEDSTOCK CREATION

Two halves of the conversion

FREED PETROL-RANGE FEEDSTOCK

Naphtha + Reformate

① Naphtha steam cracker

Ethylene · Propylene · C4 (olefins)

→ MEG, polypropylene, acrylonitrile, BDO

② Catalytic reformer + aromatics

Paraxylene · Benzene (BTX aromatics)

→ PTA → polyester, caprolactam → nylon

Synergy: the steam cracker (olefins) and aromatics complex share feed & utilities — integrating both extracts maximum value per barrel.

VALUE-ADDED PRODUCTS

Textile petrochemicals India can produce & import-substitute

Fibre	Petrochemical	Feedstock	Import vol / yr	Import dep.
Polyester (PET)	PTA (←paraxylene) + MEG	xylenes / ethylene	MEG ~1.5–2.0 MMT	partial
Acrylic (PAN)	Acrylonitrile (ACN)	propylene	~0.16 MMT (164 kt)	~100%
Nylon 6 / 6,6	Caprolactam / adipic	benzene → cyclohexane	~0.15 MMT (150 kt)	high
Spandex / PBT	BDO → PTMEG + MDI	C4 + benzene	~0.05–0.10 MMT	high
PP fibre	Polypropylene (fibre grade)	propylene	~1.6 MMT (all-use)	~20–25%
PE tape	HDPE	ethylene	~3.1 MMT (all-use)	~25–40%

The prize

4.52 MMT

India synthetic-fibre market, 2024 (PFY 2.53 MMT)

~2.0–2.4 MMT/yr

core fibre-monomer imports open to substitution (~$1.5 bn) + fibre grade of PE/PP

Cellulosic (viscose/lyocell) is wood-pulp — NOT on this petro loop.

CLOSING THE LOOP

Fuel → Chemicals → Fibre → Recycle → Fibre

1	2	3	4	5
Fuel	**Feedstock**	**Chemicals**	**Fibre**	**Recycle**
Ethanol displaces petrol (E20/E30)	Freed naphtha + reformate	Cracker + reformer → olefins & aromatics	Polyester · acrylic · nylon · spandex · PP/PE	rPET supplements virgin MMF

↺ *virgin domestic polymer + recycled polyester feed the same fibre demand*

Substitutes MMF feedstock

~2 MMT/yr of MEG, ACN, caprolactam, BDO — turns dollar imports into domestic feedstock

Closes the technical-textile gap

Feeds Mobiltech (nylon), Hometech (PP nonwoven) & specialty (PAN→carbon fibre) deficits

Captures petrochemical margin

+$1.5–2 / bbl vs thin fuel margin; virgin polymer that rPET then supplements

THE TAKEAWAY

One integrated opportunity

1	**Demand is set**	World fibre is ~75% MMF and widening; India targets a 60% MMF mix — the pull is structural, not cyclical.
2	**The gap is specialty-MMF**	India's technical-textile surplus is Packtech-only; the deficits (Mobiltech, Hometech, specialty) are all MMF/specialty-fibre.
3	**Ethanol supplies the feedstock**	E20/E30 free naphtha & reformate — the exact building blocks of polyester, acrylic, nylon, spandex, PP/PE.
4	**The loop closes**	Fuel → chemicals → fibre → rPET → fibre: substitute ~2 MMT/yr of imports and capture petrochemical margin.

Sources: DGCIS / NTTM Compendium 2024 · MATEXIL · Textile Exchange 2025 · DCPC / ICIS petrochemical tree · omc-retail-profitability-model